SHUBHAM SHRIRANG GHODEKAR

AI/ML Engineer | RAG & LLM Agent Developer | Python Backend Developer

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in Generative AI, Retrieval-Augmented Generation (RAG), and agentic workflows. Experienced in building document intelligence applications, custom MCP servers, and AI automation using Python, FastAPI, LangChain, LangGraph, Langflow and ChromaDB. Currently contributing to AI engineering projects at Techstalwarts, including PDF data extraction and LLM cost optimization.

TECHNICAL SKILLS

Generative AI & LLMs: RAG, LangChain, Prompt Engineering, Vision-Language Models (VLMs), Ollama, LLM APIs (Gemini, OpenAI, Groq)

Agentic AI & MCP: LangGraph, Langflow (Custom Components), Multi-Agent Pipelines, Model Context Protocol (MCP), FastMCP

Databases & Embeddings: ChromaDB (Vector DB), MySQL, SQLite, Embedding Models, Semantic Search

Machine Learning & Data: Scikit-learn, Ensemble Learning, Feature Engineering, Model Evaluation, NLP, Pandas, NumPy

Programming Languages: Python, JavaScript

Backend & APIs: FastAPI, Flask, REST APIs, Pydantic, Pytest

Frontend: React.js, Vite, Tailwind CSS, HTML, CSS

Tools & Automation: Docker, Git, GitHub Actions (CI), Playwright (Browser Automation), OmniRoute (LLM Gateway), PyMuPDF (PDF Parsing)

PROFESSIONAL EXPERIENCE

Techstalwarts

Aug 2026 – Present

AI/ML Intern

- Contributed to the company's AI transformation initiative by building RAG and multi-agent pipelines; applied prompt engineering, parameter tuning, and LLM APIs (Groq, OpenAI, and others) to improve retrieval and inference quality.
- Configured OmniRoute as a central LLM routing gateway and applied additional token-optimization methods to reduce the organization's overall LLM token costs.

PDF Field Extractor — Client Project (Sliceal, Dubai)

Python | FastAPI | PyMuPDF | Ollama

- Built a web-based AI application for an international client that converts PDFs and images into structured JSON for user-defined fields (e.g., invoice number, owner name, amount) — no templates or training needed.
- Implemented two extraction pipelines: a fully local path (PyMuPDF at 150 DPI → GLM-OCR via Ollama → LLM field extraction) and a direct vision-LLM path.
- Designed a provider-agnostic LLM layer for any OpenAI-compatible endpoint (Groq, OpenAI, Gemini, OpenRouter, vLLM) with FastAPI endpoints, live progress tracking, auto-managed Ollama setup, and a drag-and-drop UI.

Custom MCP Server for a Proprietary Platform

Python | FastMCP | MCP | REST APIs

- Built a custom Model Context Protocol (MCP) server with FastMCP for a proprietary software product with no native MCP support, exposing its functionality as tools that AI agents and LLM clients can call directly.
- Engineered per-endpoint context management, scoping each tool's inputs, outputs, and returned data so agents receive only the context relevant to each call.
- Defined clear tool schemas and descriptions for every endpoint, making a previously closed system accessible to agentic workflows.

Placement Co-Pilot — Vision-Enabled Browser Automation

Python | Playwright | VLM

- Configured a vision-enabled VLM with Playwright browser automation for the HR department, allowing the agent to see and navigate web pages like a human user.
- Automated lead capture and job searching across websites, reducing manual browsing and data entry for the HR team.

Agentic Workflow Automation with Langflow *Langflow | Python | Agentic AI*

- Automated repetitive internal department tasks using Langflow agentic workflows, reducing manual effort across teams.
- Developed custom Langflow nodes in Python to extend the platform with organization-specific integrations and logic.

Web-RAG — Website-to-Knowledge-Base RAG System *Python | FastAPI | Langflow | ChromaDB | Ollama*

- Built an end-to-end RAG system that converts website and documentation URLs into queryable knowledge bases using depth-controlled recursive crawling, semantic chunking, local Ollama embeddings, and ChromaDB.
- Implemented multi-tenant isolation using separate vector collections per source, integrated with Langflow via the Tweaks API to support multiple knowledge bases in a single pipeline with cited Gemini answers.
- Developed a FastAPI orchestrator with Pydantic models, multi-turn session tracking, per-request Gemini model/key overrides, and a vanilla JavaScript chat interface.
- **GitHub:** github.com/Shubham302004/web-rag

New Age Solutions Technologies (NASTECH) — Microsoft Partner Jul 2022 – Aug 2022

Mobile Application Development Intern

- Built UI screens, navigation flows, and application logic for cross-platform mobile apps using Flutter and Dart.

PERSONAL PROJECTS

Smart Resume Screener — Hybrid LLM Ranking Engine *Python | FastAPI | Gemini | React*

- Designed a hybrid resume-ranking engine combining Google Gemini for qualitative candidate evaluation with a deterministic 214-skill, 308-alias lexicon for reproducible scoring.
- Implemented weighted candidate ranking across skills, experience, education, and summary relevance, with date-range merging to prevent double-counting experience.
- Built PDF/DOCX resume parsing and a React + Tailwind dashboard featuring candidate grades, skill-gap analysis, and CSV export.
- **GitHub:** github.com/Shubham302004/Smart-Resume-Screener

Webpage Fraud Detection — Stacking Ensemble (Major Project) *Python | Scikit-learn | Flask | React*

- Developed a hybrid stacking ensemble (Random Forest + Decision Tree → Logistic Regression) using 111 URL features across 129,698 URLs, achieving 99.0% accuracy and 98.8% F1 score.
- Designed a three-stage inference pipeline combining allowlisting, heuristic filtering, and ML-based detection, with an SSRF-safe feature extractor.
- Deployed a Flask REST API and React + Tailwind dashboard featuring real-time phishing risk scores and interactive analytics, with input validation and request-level safeguards.
- **GitHub:** github.com/Shubham302004/Precise-And-Enhanced-Webpage-Fraud-Detection-Using-Machine-Learning

SMS Spam Detection — NLP Classification & API *Python, Scikit-learn, Flask, PostgreSQL, Docker, GitHub Actions*

- Developed a Multinomial Naive Bayes classifier on the UCI SMS dataset, achieving 98.57% accuracy and 0.9415 F1 score using 5-fold stratified cross-validation.
- Built a Dockerized Flask REST API with PostgreSQL connection pooling, input sanitization, and 35 automated Pytest cases integrated with GitHub Actions CI.

EDUCATION

University of Mumbai — *B.E. in Computer Science Engineering (AI & ML)* 2023 – 2026

Maharashtra State Board of Technical Education (MSBTE) — *Diploma in Information Technology* 2020 – 2023

CERTIFICATIONS

- Research Paper Presentation (Major Project) — IEEE ACROSET 2026
- Oracle Cloud Infrastructure (OCI) AI Foundations Associate — Oracle
- Build Your Own Generative AI Model — NxtWave